Nophi Ian Biton

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Education

Ph.D. – Civil Engineering Major in Disaster Management Engineering Sept. 2020 – August 2024
Ulsan National Institute of Science and Technology (UNIST) *Ulsan, South Korea*
Dissertation title: System-reliability-based design optimization of truss structures considering fatigue-induced sequential failure

M.S. in Civil and Construction Engineering June 2015 – June 2017
National Taiwan University of Science and Technology (Taiwan Tech) *Taipei, Taiwan*
Thesis title: Reliability-based design optimization using methods of moments

B.S. in Civil Engineering June 2008 – May 2013
University of the Philippines - Diliman *Quezon City, Philippines*
Thesis title: Constitutive relations of springs to model and simulate reinforced concrete frames

Research experience

UNIST Dept. of Civil, Urban, Earth and Environmental Engineering Ulsan, South Korea
My research focuses on developing a probabilistic optimization framework that considers system-level sequential failure of truss structures under fatigue-limit states. My work combines both gradient-based and gradient-free structural optimization techniques with failure-path structural system reliability analysis to satisfy target reliability thresholds.

- Developed and numerically implemented an efficient system-reliability-based design optimization framework that combines the modified branch-and-bound method and grey wolf optimization to minimize fatigue-induced structural failure, optimize material allocation, and ensure the failure probability remains below the target threshold.
- Proposed a gradient-based method for the structural optimization of trusses against sequential fatigue failure. The sensitivity of the failure probability with respect to the design variables was formulated based on the identified dominant failure sequences.
- Formulated a multi-objective structural optimization framework that considers two types of engineering uncertainty–epistemic and aleatory–using metaheuristics and Bayesian inference.

Taiwan Tech Dept. of Civil and Construction Engineering Taipei, Taiwan
My research has focused on integrating the method of moments as an efficient structural reliability analysis technique into a reliability-based design optimization framework. The proposed method was comprehensively tested on linear, highly nonlinear, and implicit performance functions.

- Developed a computational platform that links SAP2000 and Matlab to perform reliability analysis and optimization of steel frame structures, considering nonlinear analysis.
- Numerically investigated the efficiency and accuracy of a new reliability-based design optimization method that integrates the estimated failure probability using the statistical moments of a limit state function to particle swarm optimization.

Nationwide Operational Assessment of Hazards (Project NOAH) Quezon City, Philippines
My research contribution to the disaster risk reduction research program funded by the Philippine government involves modeling and simulating storm surge events using Delft3D software for hazard mapping. I conducted fieldwork validation along the Philippine coastline and monitored near-real-time storm surge levels during a typhoon.

Research Interest

My broad research interests are:

- Structural System Reliability
- Reliability-based design optimization
- Multidisciplinary optimization
- Probabilistic analysis for earthquake engineering and disaster risk reduction studies

Scholarship Awards

Ulsan National Institute of Science and Technology Scholarship (Korean Government)
Ministry of Science and Technology (MOST) Scholarship (Taiwan Government)
Megaworld Foundation Scholarship
Department of Science and Technology (DOST) PSHS Scholarship (Philippine Government)

Teaching Experience

Far Eastern University – Institute of Technology Manila, Philippines
Lecturer *Sept. 2021 – April 2022*
Delivered review lectures for the Philippine Civil Engineering Licensure Examination. Created an online question bank for practice assessments. Taught numerical and structural analysis courses using Matlab programming software

University of San Carlos – Technological Center Cebu City, Philippines
Lecturer *Nov. 2017 – May 2020*
Taught undergraduate courses such as Statics of Rigid Bodies, Mechanics of Deformable Bodies, and Structural Theory. Developed course syllabi and prepared homework assignments, comprehensive exams, and solutions. Delivered lectures for graduate courses such as Finite Element Method and Structural Dynamics. Developed final projects for students. Reviewed lecture material and solved example problems.

Advising Experience

University of San Carlos - Technological Center Cebu City, Philippines
Research Adviser *Nov. 2017 – May 2020*
Advised undergraduate students on their thesis requirements for a bachelor's degree. The research focused on the structural optimization of standard government school buildings in the Philippines.

Industry Experience

Rapid Forming Corporation San Juan Del Monte, Philippines
Design Engineer *Mar. 2015 – Jun 2015*
Worked as part of the structural design team at Rapid Forming Corporation, which distributes, designs, and builds frame structures made of cold-formed steel. The work involved detailed structural design, bill of quantities, and cost estimation.

Scientific Articles

- **Biton, N. I.**† & Lee, Y.-J. (For submission). Multi-objective system reliability-based design optimization considering mixed aleatory and epistemic uncertainty on fatigue-induced sequential failures of trusses.
- **Biton, N. I.**†, Kang, W.-H., Chun, J., & Lee, Y.-J. (2024). Probabilistic failure path approach on optimal design of structures against sequential fatigue-induced failure, Structural and multidisciplinary optimization, 67. https://doi.org/10.1007/s00158-024-03918-4
  Peer Reviewed | Impact Factor: 3.6 | **Quartile (SJR)**: Q1
- **Biton, N. I.**† & Lee, Y.-J. (2024). Structural system reliability-based design optimization considering fatigue limit state, Smart Structures and Systems. https://doi.org/10.12989/sss.2024.33.3.177
  Peer Reviewed | Impact Factor: 2.1 | **Quartile (SJR)**: Q2
- Liao, K. W., & **Biton, N. I.**† (2020). A heuristic moment-based framework for optimization design under uncertainty. Engineering with Computers, 1-14. https://doi.org/10.1007/s00366-019-00759-4
  Peer Reviewed | Impact Factor: 8.7 | **Quartile (SJR)**: Q1
- Liao, K. W., & **Biton, N. I.**† (2019). A heuristic optimization considering probabilistic constraints via an equivalent single variable Pearson distribution system. Applied Soft Computing, 78, 670-684. https://doi.org/10.1016/j.asoc.2019.03.021
  Peer Reviewed | Impact Factor: 7.2 | **Quartile (SJR)**: Q1

Conference Proceedings

- **Biton, N. I.**† & Lee, Y.-J. (2024, September). Bayesian approach for addressing epistemic and aleatory uncertainty in the optimal fatigue design. In 8th Asia-Pacific Symposium on Structural Reliability and Its Applications (APSSRA 2024). Ulsan, South Korea. [Oral presentation]
- **Biton, N. I.**† & Lee, Y.-J. (2023, August).Structural system reliability-based design optimization considering fatigue limit state. In the 2023 World Congress on Advances in Structural Engineering and Mechanics (ASEM23). Seoul, South Korea. [Oral presentation]
- **Biton, N. I.**† & Lee, Y.-J. (2023, July). Probabilistic failure path approach on structural system-reliability-based design optimization of fatigue-induced failure. In 14^{th} International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP14). Dublin, Ireland. [Oral presentation]
- **Biton, N. I.**† & Lee, Y.-J. (2022, October). Reliability-based Design Optimization considering Fatigue limit state. In Korean Society of Civil Engineers Convention 2022. Busan, South Korea. [Oral presentation]
- Camanse, S. S., **Biton, N. I.**† & Giduquio, M. B. (2019, October). Properties of Concrete using Fly ash as an Aggregate Micro-filler. In 16^{th} Engineering Research and Development for Technology Conference. Manila, Philippines. [Poster presentation]
- Llanita, M. C., **Biton, N. I.**† & Giduquio, M. B. (2019, October). Mechanical Properties of Moderate-Slump Concrete Containing High-Volume Fly Ash. In 16^{th} Engineering Research and Development for Technology Conference. Manila, Philippines. [Poster presentation]
- Suarez, J. K., Cabacaba, K. M., **Biton, N. I.**†, Cuadra, C., Santiago, J., Mendoza, J., & Mahar Francisco Lagmay, A. (2015, April). Delft3D Storm Surge Simulation of Typhoon Haiyan for Projection of Coastal Inundation in the Visayas Islands, Philippines. In EGU General Assembly Conference Abstracts (Vol. 17). [Poster presentation]
- Cuadra, C., Suarez, J. K., **Biton, N. I.**†, Cabacaba, K. M., Lapidez, J. P., Santiago, J., & Malano, V. (2014, May). Development of Inundation Map for Bantayan Island, Cebu Using Delft3D-Flow Storm Surge Simulations of Typhoon Haiyan. In EGU General Assembly Conference Abstracts (Vol. 16). Sapporo, Japan. [Poster presentation]

Technical Skills

Programming Languages: Matlab, Python
Structural Software: SAP2000, ETABS, SAFE, Xtract, OpeenSeesPy
Spoken Language: Cebuano (Native), Filipino (Fluent), English (Fluent)

Training, Workshops and Seminars

- SAP2000 Structural Analysis and Design
- Association of Structural Engineers of the Philippines – Design of Steel and RC SMRF
- Machine Learning, Neural Networks and Deep Learning
- Course on Optimization problems and Algorithms

Hobbies and interests

I enjoy exploring national parks and engaging in hiking and biking over shorter distances. I am interested in further expanding my skills in terms of scientific programming, machine learning and structural reliability. I share my scripts and personal notes at https://github.com/nophibiton .